Questions:

1. What is a magnet?

2. Name two magnetic materials.

3. What happens when two like poles (for example, north and north) are placed near each other?

4. What is the difference between a magnetically hard material and a magnetically soft material?

5. What is a magnetic field?

6. How can we make a magnetic field visible?

7. What is induced magnetism?

8. What is an electromagnet?

9. State the Right-Hand Grip Rule.

10. List two ways to make a solenoid stronger.

Answer key:

1. A magnet is an object that attracts magnetic materials such as iron, steel, nickel, and cobalt.

2. Iron and steel. (Nickel or cobalt are also correct.)

3. They repel each other.

4. Magnetically hard materials keep their magnetism permanently, while magnetically soft materials lose their magnetism easily.

5. A magnetic field is the invisible area around a magnet where magnetic force can be detected.

6. By using iron filings.

7. Induced magnetism is the process of turning a magnetic material into a magnet by placing it near or touching a magnet.

8. An electromagnet is a magnet made using electricity.

9. The Right-Hand Grip Rule is a method to find the direction of the magnetic field using your right hand, with the thumb pointing to the north direction.

10. Increase the current and increase the number of turns of the coil. (Adding an iron core is also correct.)

1. A speaker that vibrates makes a sound wave in the air with a wavelength of 34 m. If the speed of sound waves is 340 m/s, then what is the frequency of the sound?

⊕ Known :

$\lambda = 34m$

$v = 340 m/s$

⊕ Wanted :

$f = ?$

⊕ Answer :

$v = \lambda . f$

$\frac{340}{34} = \boxed{34} f$

$f = 10 Hz$

2. A source of vibration produces 600 vibrations in 1 minute. Calculate the wavelength produced if the speed of waves in the air is 340 m/s?

⊕ Known :

$n = 600$

$t = 1 min$

$= 60 s$

$v = 340 m/s$

⊕ Wanted

$\lambda = ?$

⊕ Answer

→ $v = \lambda . f$ → find this first.

→ $f = \frac{n}{t} = \frac{600}{60} = 10 Hz$

→ $v = \lambda . f$

$\frac{340}{10} = \lambda . \boxed{10}$

$\lambda = \frac{340}{10} = 34 m$

3. What is the period and speed of a wave for a wave that has frequency 2 kHz and wavelength 17 cm?

⊕ Known

$f = 2 kHz = 2000 Hz$

$\lambda = 17 cm = \frac{17}{100} m$

⊕ Wanted :

$T = ?$

$v = ?$

⊕ Answer:

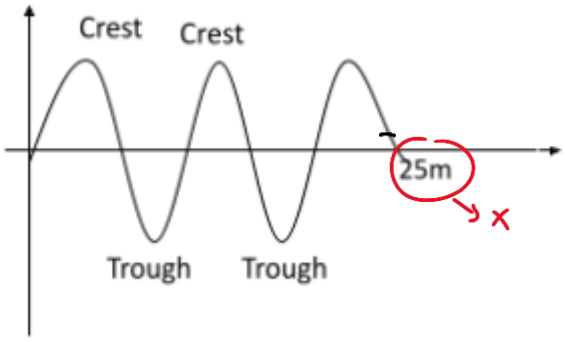
→ $T = \frac{1}{f} = \frac{1}{2000} s$

→ $v = \lambda . f$

$= \frac{17}{\cancel{100}} . 20\cancel{00}$

$v = 340 m/s$

4. The picture shows a wave that travel at 80 m/s
   a. What is the wavelength of the wave? → $\lambda = ?$
   b. What is the frequency of the wave? → $f = ?$



• Known:

$v = 80\,m/s$ $\quad n = 2.5$

$x = 25\,m$

• Wanted:

$\lambda = ?$

$f = ?$

• Answer:

→ $n.\lambda = x$

$\boxed{2.5}.\lambda = \frac{25}{2.5}$

$\lambda = 10\,m$

→ $v = \lambda . f$

$80 = 10 . f$

$f = 8\,m/s$

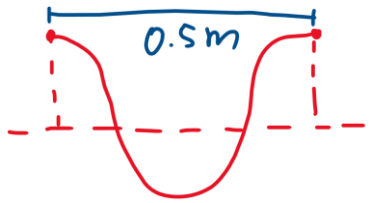
5. A tsunami wave travels in the ocean, from the measurement, it is able to travel 350 m in 5 s. What is the frequency and period of the wave if the distance between two tops of waves is 0.5 m?

• Known:

$x = 350\,m$ → considered as distance travelled by the wave

$t = 5\,s$

$\lambda = 0.5\,m$ → distance between 2 tops



• Wanted:

$f = ?$ $\quad T = ?$

• Answer:

$v = \lambda . f$

find this first

• $v = \frac{x}{t} = \frac{350}{5} = 70\,m/s$

⇒ $v = \lambda . f$

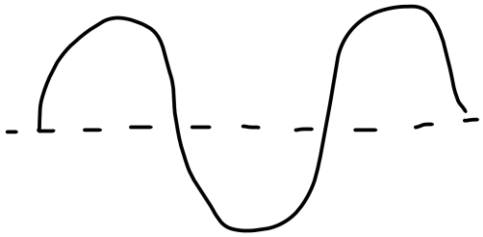
$\frac{70}{0.5} = \boxed{0.5} . f$

$f = 140\,Hz$

⇒ $T = \frac{1}{f} = \frac{1}{140}\,s$

6. 2 crests and 1 trough of waves are produced in 1.5 s.
   a. What is the frequency of the wave?
   b. What is the period of the wave?

• Known :



2 crests and 1 trough ⇒ $n = 1.5$

$t = 1.5\ s$

• Wanted :

$f = ?$ $T = ?$

• Answer

$f = \frac{n}{t} = \frac{1.5}{1.5} = 1\ Hz$

$T = \frac{1}{f} = 1\ s$

7. A piece of cork floats above sea water. The cork moves upside down 100 times in 40 s. In the same time the sea wave has traveled 60 m. What is the wavelength of the wave?

• Known :

$n = 100$ → 1 upside down means 1 vibration / wave

$t = 40\ s$

$x = 60\ m$

• Wanted :

$\lambda = ?$

• Answer :

$n . \lambda = x$

$100 . \lambda = 60$

$\lambda = \frac{60}{100} = 0.6\ m$